

Arguments, Logic & Fallacies

24 practice flashcards in four sets, a spaced-review schedule and a mixed practice quiz

■ Set A: Argument basics	cards 1-6
■ Set B: Validity, strength & first fallacies	cards 7-12
■ Set C: Informal fallacies, part 1	cards 13-18
■ Set D: Informal fallacies, part 2 & biases	cards 19-24

Schedule Each set is reviewed 1, 3, 7 and 14 days after you learn it. Tick each box when done.

DAY	TASK	CARDS	DATE	DONE
1	Learn Set A	6	_____	☐
2	Learn Set B, review Set A	12	_____	☐
3	Learn Set C, review Set B	12	_____	☐
4	Learn Set D, review Sets A and C	18	_____	☐
5	Review Sets B and D	12	_____	☐
6	Review Set C	6	_____	☐
7	Review Set D	6	_____	☐
8	Review Set A	6	_____	☐
9	Review Set B	6	_____	☐
10	Review Set C	6	_____	☐
11	Review Set D	6	_____	☐
15	Final review: Set A	6	_____	☐
16	Final review: Set B	6	_____	☐
17	Final review: Set C	6	_____	☐
18	Final review: Set D	6	_____	☐

How to use it

- **Answer first.** Say or write your answer before flipping the card. Recalling it is what builds memory, more than rereading.
- **Missed cards restart.** If you get a card wrong, treat it as new and restart its intervals.
- **Keep sessions short.** Most days have 6-12 cards (about 10-15 minutes). Day 4 is the heavy day, with 18 cards, so allow about 20 minutes.
- **Check module-specific cards.** Cards 15 (begging the question), 20 (naturalistic fallacy) and 21 (false analogy) should be checked against your course materials.
- **Printing.** Print double-sided, **flip on long edge**, then cut along the dashed lines. Each answer lands on the back of its question.

Back of the schedule. Left blank so the cards line up when printed double-sided.

SET A #1

answer before you flip

What is the difference between an **opinion** and a **philosophical argument**?

SET A #2

answer before you flip

Name the **two parts** of every argument and say what each one does.

SET A #3

answer before you flip

List three **premise indicators** and three **conclusion indicators**.

SET A #4

answer before you flip

What is the difference between **truth value analysis** and **logical analysis** of an argument?

SET A #5

answer before you flip

"The pavement is wet **because** it rained last night."

Is this an **argument** or an **explanation**?
How can you tell?

SET A #6

answer before you flip

What is the difference between a **deductive** and an **inductive** argument?

#2 ANSWER

Premises and conclusion

Premises: the statements offered as reasons or evidence.

Conclusion: the claim the premises are meant to support.

All humans are mortal. Socrates is human. (premises)
So Socrates is mortal. (conclusion)

#1 ANSWER

Opinion vs. philosophical argument

An **opinion** is a claim stated without support. A **philosophical argument** gives reasons (premises) meant to support a claim (conclusion), so it can be checked and judged.

"*Lying is wrong*" is an opinion. Adding *why* it is wrong turns it into an argument.

#4 ANSWER

Truth value vs. logical analysis

Truth value analysis asks: are the premises actually true?

Logical analysis asks: *if* the premises were true, would the conclusion follow?

A good argument passes both tests.

#3 ANSWER

Indicator words

Premise indicators: because, since, given that, for, as shown by, the reason is.

Conclusion indicators: therefore, thus, so, hence, consequently, it follows that.

They are clues, not guarantees. Always check the role each sentence plays.

#6 ANSWER

Deductive vs. inductive

Deductive: claims the conclusion follows *necessarily*. If the premises are true, the conclusion must be true.

Inductive: claims the conclusion is *probably* true given the premises.

Every swan I have seen is white, so the next one will be is inductive.

#5 ANSWER

Argument vs. explanation

Explanation. Nobody doubts that the pavement is wet; the sentence says *why* it is.

An **argument** tries to show *that* a claim is true. An **explanation** says *why or how* something already accepted is true. "Because" can appear in both.

SET B #7

answer before you flip

What is the difference between a **valid** and a **sound** argument?

SET B #8

answer before you flip

How does the **counterexample method** show that an argument is invalid?

SET B #9

answer before you flip

What is the difference between a **strong** and a **cogent** inductive argument?

SET B #10

answer before you flip

What is an **enthymeme**, and what does the **principle of charity** tell you to do with one?

SET B #11

answer before you flip

What is the difference between a **formal** and an **informal** fallacy?

SET B #12

answer before you flip

Name the fallacy.

"Don't listen to her plan for the city budget.
She's been divorced twice."

#8 ANSWER

Counterexample method

Build an argument with the **same form** where the premises are clearly true and the conclusion is clearly false. If that can happen, the form is invalid.

All cats are animals. All dogs are animals. So all cats are dogs.

#7 ANSWER

Valid vs. sound

Valid: it is impossible for the premises to be true and the conclusion false. This is about form, not actual truth.

Sound: valid *and* every premise is actually true.

A valid argument can have false premises. A sound argument cannot.

#10 ANSWER

Enthymeme and principle of charity

Enthymeme: an argument with an unstated premise (or conclusion).

Principle of charity: rebuild other people's arguments in their strongest, most reasonable form, filling gaps with the most plausible missing premise.

"He's a lawyer, so he's good at arguing" assumes lawyers are good at arguing.

#9 ANSWER

Strong vs. cogent

Strong: if the premises were true, the conclusion would probably be true.

Cogent: strong *and* the premises are actually true, with no key evidence left out.

Strong/cogent is the inductive version of valid/sound.

#12 ANSWER

Ad hominem

Attacking the **person** (their character, circumstances or hypocrisy) instead of their argument.

Her divorces say nothing about whether the budget plan is any good.

#11 ANSWER

Formal vs. informal fallacies

Formal: an error in the argument's *structure*, so you can spot it from the form alone (e.g. affirming the consequent: if P then Q; Q; so P).

Informal: an error in the *content*, such as irrelevant reasons, weak evidence or hidden assumptions. You have to look at what is said.

SET C #13

answer before you flip

Name the fallacy.

"I met two people from that town and both were rude. Everyone there must be rude."

SET C #14

answer before you flip

Name the fallacy.

"If we let students retake one test, soon they'll want to retake every test, and then grades will mean nothing."

SET C #15

answer before you flip

Name the fallacy.

"This news source is trustworthy because it never reports anything false."

SET C #16

answer before you flip

Name the fallacy.

"A famous actor says this diet cures diabetes, so it must work."

SET C #17

answer before you flip

Name the fallacy.

"No one has ever proven ghosts don't exist, so they must be real."

SET C #18

answer before you flip

Name the fallacy.

"Either you support this law, or you don't care about children's safety."

#14 ANSWER

Slippery slope

Claiming one step will lead through a **chain of events** to an extreme outcome, with no good reason to think each link will actually happen.

#13 ANSWER

Hasty generalization

Drawing a general conclusion from a sample that is **too small or unrepresentative**.

Two people cannot represent a whole town.

#16 ANSWER

Appeal to (unqualified) authority

Treating someone's word as proof when they are **not an expert** on the topic (or when experts disagree).

Citing a relevant expert consensus is *not* a fallacy.

#15 ANSWER

Begging the question

The premises **assume the conclusion** they are meant to prove (circular reasoning). Whether it never reports anything false is exactly the question.

Check this definition against your course materials.

#18 ANSWER

False dichotomy

Presenting **only two options** when others exist. You could care about safety and still think this law is a bad way to get it.

#17 ANSWER

Appeal to ignorance

Concluding a claim is true because it has not been **proven false** (or false because it has not been proven true). A lack of evidence is not evidence.

SET D #19

answer before you flip

Name the fallacy.

A: "We should cut the military budget a bit."

B: "So you want to leave the country defenceless?"

SET D #20

answer before you flip

Name the fallacy.

"Herbal remedies are natural, so they must be better for you than medicine."

SET D #21

answer before you flip

Name the fallacy.

"Employees are like nails. Nails have to be hit on the head to work, so employees do too."

SET D #22

answer before you flip

Name the **bias**: someone sure a stock will rise reads only the bullish news and dismisses every warning.

SET D #23

answer before you flip

Name the **bias**: after watching news about a plane crash, someone decides flying is more dangerous than driving.

SET D #24

answer before you flip

Name the **effect**: people with the least skill at a task are often the most confident they are good at it.

#20 ANSWER

Naturalistic fallacy

Concluding that something is good or right **because it is natural**, or more broadly deriving an "ought" from an "is".

Check this definition against your course materials, since courses frame it differently.

#19 ANSWER

Straw man

Misrepresenting an opponent's position as a weaker or more extreme version, then attacking that version instead of the real one.

#22 ANSWER

Confirmation bias

The tendency to **seek out, notice and remember** evidence that supports what we already believe, and to ignore or discount evidence against it.

#21 ANSWER

False analogy

Arguing that two things alike in some ways must be alike in another, when the **similarities are irrelevant** or the differences matter.

Check this definition against your course materials.

#24 ANSWER

Dunning-Kruger effect

People with **low ability** in an area tend to **overestimate** their competence, partly because they lack the skill to see their own mistakes.

#23 ANSWER

Availability error

Judging how likely or common something is by **how easily examples come to mind**. Vivid or recent events feel more frequent than they are.

Mixed practice quiz

Use after Day 11. Name the fallacy or bias, or judge the argument. Answer key on the next page.

1. "My uncle smoked all his life and lived to 95, so smoking can't be that bad."

Answer: _____

2. "You can't trust his argument about climate policy; he drives an SUV."

Answer: _____

3. "Nobody has shown this supplement is harmful, so it's safe."

Answer: _____

4. "We either ban phones in class entirely or accept that no one will learn."

Answer: _____

5. "Free will exists, because we are free to choose what we do."

Answer: _____

6. Feminists want equal pay. Critic: "They want to make men second-class citizens."

Answer: _____

7. "If we allow food in the library, next it'll be parties, then it'll be trashed."

Answer: _____

8. "A Nobel-winning physicist says this vaccine is unsafe, so it is."

Answer: _____

9. "The universe is like a watch. Watches have makers, so the universe does too."

Answer: _____

10. "Animals eat meat in nature, so eating meat must be morally right."

Answer: _____

11. All birds can fly. Penguins are birds. So penguins can fly. Valid? Sound?

Answer: _____

12. If it rains, the ground is wet. The ground is wet. So it rained. Valid?

Answer: _____

Answer key

1. Hasty generalization
2. Ad hominem (tu quoque / hypocrisy)
3. Appeal to ignorance
4. False dichotomy
5. Begging the question
6. Straw man
7. Slippery slope
8. Appeal to authority (outside their field)
9. False analogy (check against your course)
10. Naturalistic fallacy
11. Valid but not sound (premise 1 is false)
12. Invalid: affirming the consequent (a formal fallacy)

Score 10+ out of 12: you're in good shape. Any you missed: find that card and restart its intervals.